



Going electric

Land

Trolley pole transfers electricity from overhead cable to train's motor



B&O Bo Switcher USA 1895

Door to engineers's cab

Pantograph collects power from overhead cable



GIPR Class WCP 1 UK/India 1930



NER Electric Locomotive UK 1905



Each half of locomotive has two motors to drive the wheels

Streamlined nose



PRR Class GG1 USA 1934



DRE04 Germany 1933

A GG1 pulled the funeral train of President Franklin D. Roosevelt.



The 1880s saw electric streetcars and trolleys rattling around cities, and it was not long before electric trains appeared. They offered advantages over smoke-belching steam trains, but they needed electrified railroad lines on which to run.

Experimental electric trains had been built since the 1830s, but the first main line electric service was in Baltimore in the 1890s. The **B&O Bo Switcher** operated in Baltimore's docklands at a top speed of 9 mph (16km/h). Electric trains get their power supply either from overhead cables or